Project Design Phase

Solution Architecture

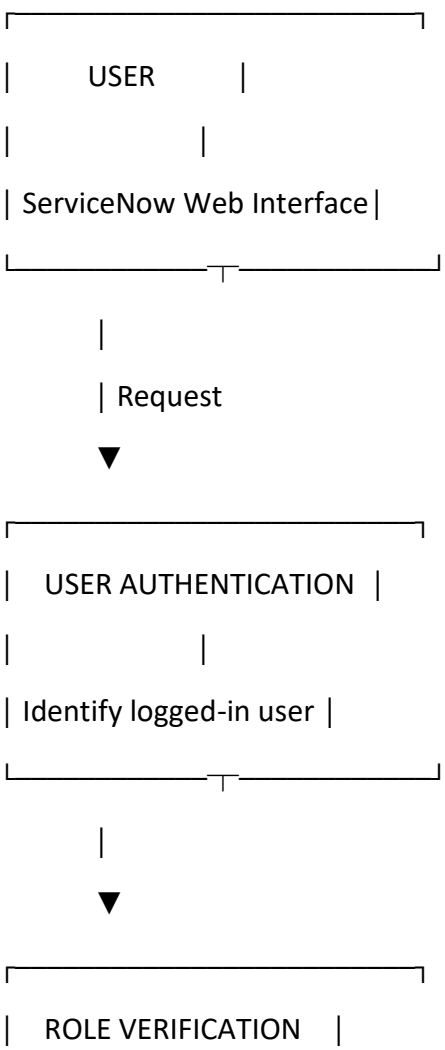
Date: 03 October 2026
Team ID: 6aba117e538da4003d63184c
Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks: 4 Marks

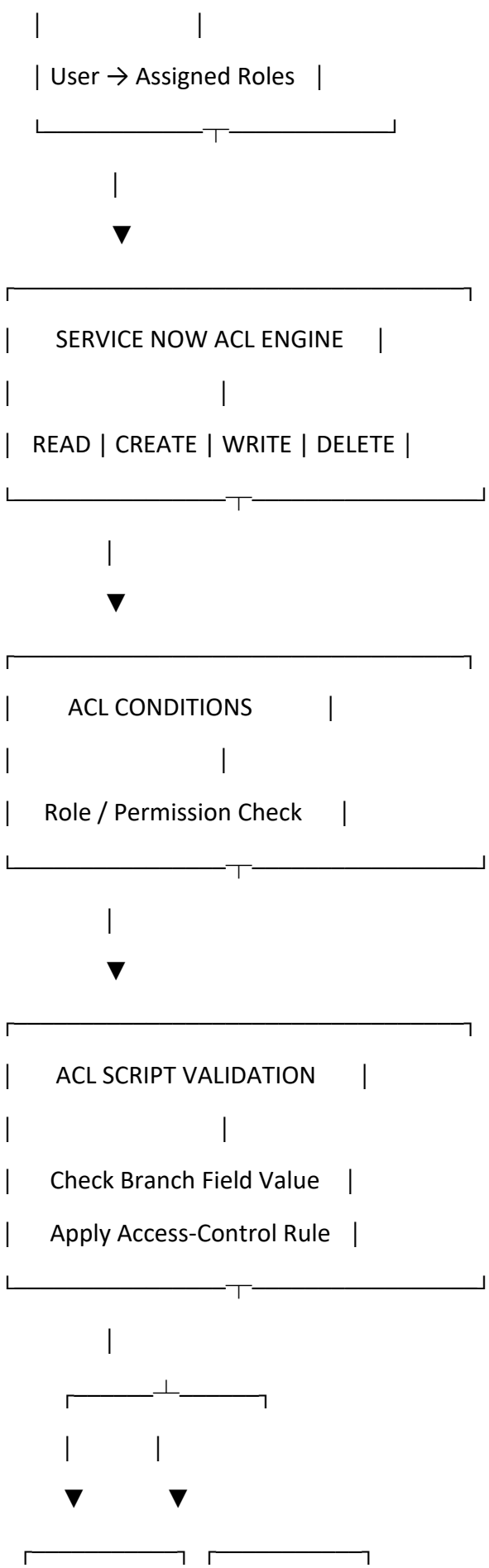
Solution Architecture

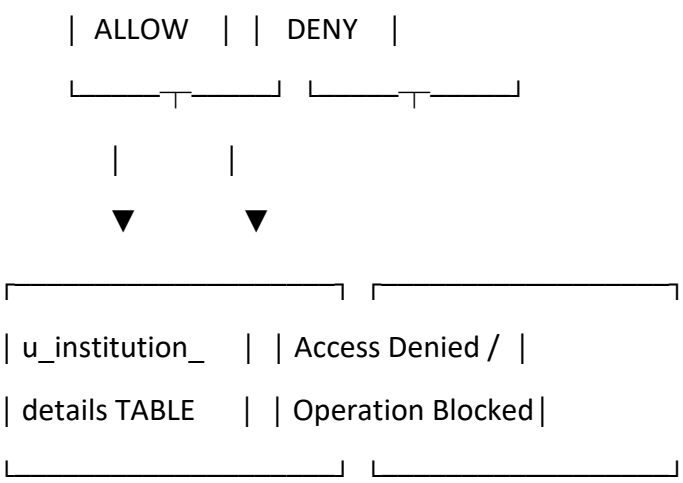
The proposed solution is implemented on the **ServiceNow platform** to provide secure and granular access control for the u_institution_details table. The architecture combines **users, roles, ACL rules, ACL conditions and server-side scripts** to determine whether a user is authorized to perform a particular operation.

When a user requests access to an institution record, ServiceNow first identifies the user's assigned role and then evaluates the applicable ACL. The ACL verifies the requested operation, such as **READ, CREATE, WRITE or DELETE**. For branch-specific restrictions, the ACL script evaluates the **Branch** field of the record. Based on these checks, the system either grants or denies the requested operation.

Figure 1: Solution Architecture and Data Flow




User → Assigned Roles
SERVICE NOW ACL ENGINE
READ | CREATE | WRITE | DELETE
ACL CONDITIONS
Role / Permission Check
ACL SCRIPT VALIDATION
Check Branch Field Value
Apply Access-Control Rule



Architecture Components

Component	Purpose
User	Initiates requests to view or modify institution records.
ServiceNow Web Interface	Provides the interface through which users interact with the system.
User Authentication	Identifies the logged-in ServiceNow user.
User & Role Management	Determines the roles and permissions assigned to each user.
ACL Engine	Evaluates access permissions for READ, CREATE, WRITE and DELETE operations.
ACL Conditions	Provides role-based and condition-based access control.
ACL Script	Performs additional validation based on the Branch field and other record information.
u_institution_details Table	Stores institution details and associated branch information.
Access Decision	Grants or denies the requested operation based on the configured security rules.

Working Flow

User Request → Authentication → Role Verification → ACL Evaluation → Condition Check → Branch Validation → Allow/Deny → Record Operation

Architectural Approach

The solution follows a **centralized access-control architecture**. All authorization decisions are handled by ServiceNow's ACL mechanism rather than implementing separate access-control logic for individual users. This makes the solution easier to administer and extend.

The architecture supports:

- **Role-based access control**
- **Record/field-based access restriction**
- **READ, CREATE, WRITE and DELETE protection**
- **Server-side script validation**
- **Centralized security management**
- **Extension to additional branches and roles**
- **Controlled access to sensitive institution records**